```java
import java.util.Objects;

public class Book {
    private final String title;
    private final String author;
    private final int pages;
    private final int year;

    public Book(String title, String author, int pages, int year) {
        this.title = title;
        this.author = author;
        this.pages = pages;
        this.year = year;
    }

    public String getTitle() {
        return title;
    }

    public String getAuthor() {
        return author;
    }

    public int getPages() {
        return pages;
    }

    public int getYear() {
        return year;
    }

    @Override
    public boolean equals(Object o) {
        if (this == o) return true;
        if (!(o instanceof Book)) return false;
        Book book = (Book) o;
        return pages == book.pages &&
                year == book.year &&
                Objects.equals(title, book.title) &&
                Objects.equals(author, book.author);
```

```
41      }
42
43      @Override
44      public int hashCode() {
45          return Objects.hash(title, author, pages,
    year);
46      }
47
48      @Override
49      public String toString() {
50          return "Book{" +
51                  "title='" + title + '\'' +
52                  ", author='" + author + '\'' +
53                  ", pages=" + pages +
54                  ", year=" + year +
55                  '}';
56      }
57  }
58
```

```java
import java.io.BufferedReader;
import java.io.IOException;
import java.io.InputStream;
import java.io.InputStreamReader;
import java.nio.charset.StandardCharsets;
import java.nio.file.Files;
import java.nio.file.Paths;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.List;
import java.util.Optional;

public class Dz2Main {

    public static void main(String[] args) throws IOException {
        List<Student> students = new ArrayList<>();

        try (BufferedReader reader = new BufferedReader(
                new InputStreamReader(openStudentsFile(), StandardCharsets.UTF_8))) {

            String name = null;
            List<Book> books = null;

            for (String line = reader.readLine(); line != null; line = reader.readLine()) {
                String trimmed = line.trim();

                if (trimmed.isEmpty()) {
                    if (name != null) {
                        students.add(new Student(name, books));
                        name = null;
                        books = null;
                    }
                } else if (trimmed.startsWith("Student:")) {
                    if (name != null) {
                        students.add(new Student(name
```

```
35 , books));
36                     }
37                     name = trimmed.substring("Student
   :".length()).trim();
38                     books = new ArrayList<>();
39                 } else if (trimmed.startsWith("Book:"
   ) && books != null) {
40                     String[] parts = trimmed.
   substring("Book:".length()).trim().split("\\|");
41                     books.add(new Book(
42                             parts[0].trim(),
43                             parts[1].trim(),
44                             Integer.parseInt(parts[2
   ].trim()),
45                             Integer.parseInt(parts[3
   ].trim())));
46                 }
47             }
48
49             if (name != null) {
50                 students.add(new Student(name, books
   ));
51             }
52         }
53
54         students.stream()
55                 .peek(System.out::println)
56                 .flatMap(student -> student.getBooks
   ().stream())
57                 .sorted(Comparator.comparingInt(Book
   ::getPages))
58                 .distinct()
59                 .filter(book -> book.getYear() > 2000
   )
60                 .limit(3)
61                 .map(Book::getYear)
62                 .reduce(Optional.<Integer>empty(),
63                         (found, year) -> found.or
   (() -> Optional.of(year)),
64                         (first, second) -> first.or
   (() -> second))
```

```
65                  .ifPresentOrElse(
66                          year -> System.out.println("
   Год выпуска найденной книги: " + year),
67                          () -> System.out.println("
   Такая книга отсутствует"));
68      }
69
70      private static InputStream openStudentsFile()
   throws IOException {
71          InputStream fromClasspath = Dz2Main.class.
   getResourceAsStream("/students.txt");
72          if (fromClasspath != null) {
73              return fromClasspath;
74          }
75          return Files.newInputStream(Paths.get("
   resources", "students.txt"));
76      }
77 }
78
```

```java
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Objects;

public class Student {
    private final String name;
    private final List<Book> books;

    public Student(String name, List<Book> books) {
        this.name = name;
        this.books = Collections.unmodifiableList(new ArrayList<>(books));
    }

    public String getName() {
        return name;
    }

    public List<Book> getBooks() {
        return books;
    }

    @Override
    public boolean equals(Object o) {
        if (this == o) return true;
        if (!(o instanceof Student)) return false;
        Student student = (Student) o;
        return Objects.equals(name, student.name) &&
                books.equals(student.books);
    }

    @Override
    public int hashCode() {
        return Objects.hash(name, books);
    }

    @Override
    public String toString() {
        return "Student{" +
                "name='" + name + '\'' +
```

```
41                  ", books=" + books +
42                  '}';
43      }
44 }
45
```